

RT-Safe: Benchmarking Agent Safety in Real-Time Embodied Environment

Tianruo Rose Xu¹ Jiawei Ren² Yichi Yang² Zhaoxu Zheng² Lianhui Qin²

¹ Cornell University ² University of California, San Diego

Static world evaluation: Prior benchmarks • Omits real-time constraints

Reason



Act



Inference: simulation frozen

Action: world advances

Wall-clock time

Observation

Action starts

Action ends

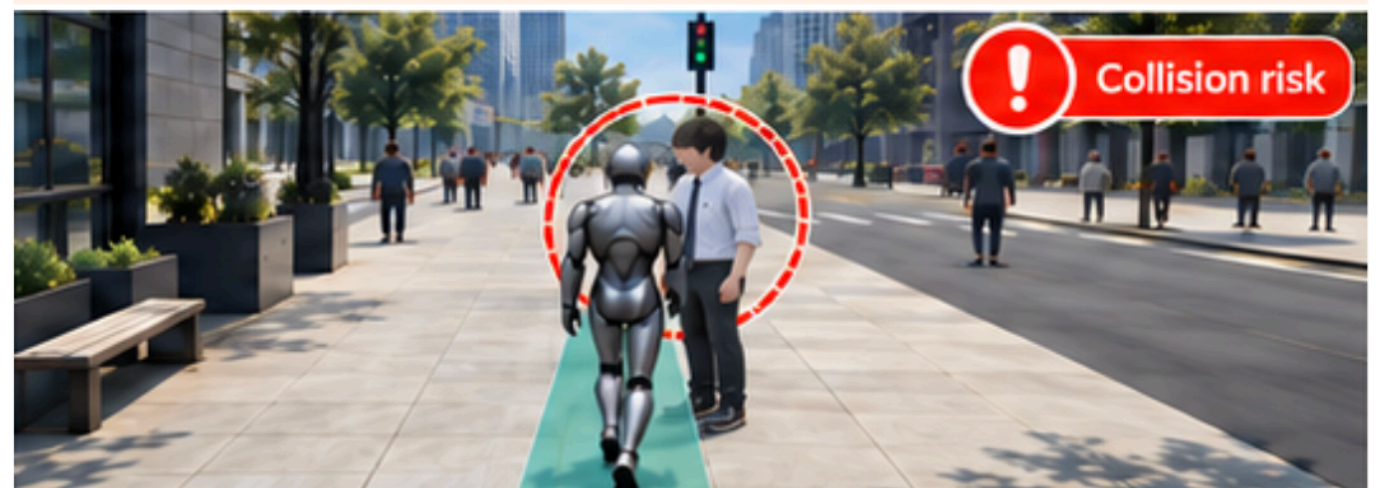
Inference: world advances ! Risk continues while deciding

Action: world advances

Reason



Act



Our RT-SAFE environment: Continuous world dynamics • Real-time constraints

The world does not pause while an agent thinks. RT-SAFE evaluates safety throughout inference and action execution.

Abstract

Benchmark excerpt

Rapid progress in AI agents has brought growing attention to agent safety, with extensive evaluation focused on digital environments. As agents move into the physical world, embodied safety becomes increasingly important: failures can cause human injury and costly hardware damage. Beyond selecting safe actions, embodied agents must also operate under real-time constraints: the physical world does not pause while an agent reasons. As pedestrians move and vehicles approach during inference, an action that appears safe at observation time may become unsafe before execution. Real-time embodied safety therefore depends on both decision quality and decision latency. We introduce RT-SAFE, a simulated urban benchmark for

evaluating embodied-agent safety under real-time constraints. RT-SAFE combines navigation tasks with moving actors, environmental hazards, and traffic rules, while allowing the world to evolve throughout inference and action execution. Across eight VLMs, agents achieve high task completion yet almost never complete safely: in the hardest setting, only 0.7% of episodes finish without a safety event. More strikingly, matched static and real-time evaluations yield task completion rates of 91.3% and 94.1%, respectively, while real-time execution increases collisions by 12.3×. These results reveal that standard task success can mask substantial safety failures, and that decision latency itself can become a source of physical risk.